

EXPERIMENT - 7

Date: 2082-10-21

TO VERIFY THE LAW OF REFRACTION OF LIGHT AND TO STUDY THE VARIATION OF LATERAL SHIFT WITH ANGLE OF INCIDENCE

APPARATUS REQUIRED:

- i. Glass slab ii. Drawing Board
 iii. Paper iv. Fixing pins, Hair pins v. Instrument box

THEORY:

The laws of refraction of light states that

- i. For a ray proceeding from the rarer to a denser medium, the sine of the angle of incidence bears to the sine of the angle of refraction a constant ratio μ given by: $\mu = \frac{\sin i}{\sin r}$. This is known as Snell's law.
 ii. The incident ray, the refracted ray and the normal to the surface at the point of incidence are in the same plane.

The lateral shift or displacement is related to the angle of incidence and the thickness of the glass slab by the relation

$$P = \frac{t \sin(i - r)}{\cos r}$$

Where,

P is the lateral shift i.e. the perp. distance between the emergent ray and the produced incident ray, t is the thickness of the slab, i and r the angle of incidence and refraction respectively.

$$\text{So, } P = \frac{t \sin(90^\circ - r)}{\cos r} = \frac{t \cos r}{\cos r} = t$$

Thus the lateral shift corresponding to $i = 90^\circ$ is equal to the thickness of given glass slab.

OBSERVATION

Least count of protector = 1°

Least count of scale = 0.1 mm

Table 1: For the determination of refractive index (μ) (using protractor)

No. of Obs.	Angles of incidence (i)	Sin i	Angles of refraction (r)	Sin r	$\mu = \frac{\sin i}{\sin r}$	Mean (μ)
1	10°	0.17	6°	0.10	1.7	1.73 =
2	20°	0.34	11°	0.19	1.78	
3	30°	0.5	17°	0.29	1.72	
4	40°	0.64	22°	0.37	1.72	

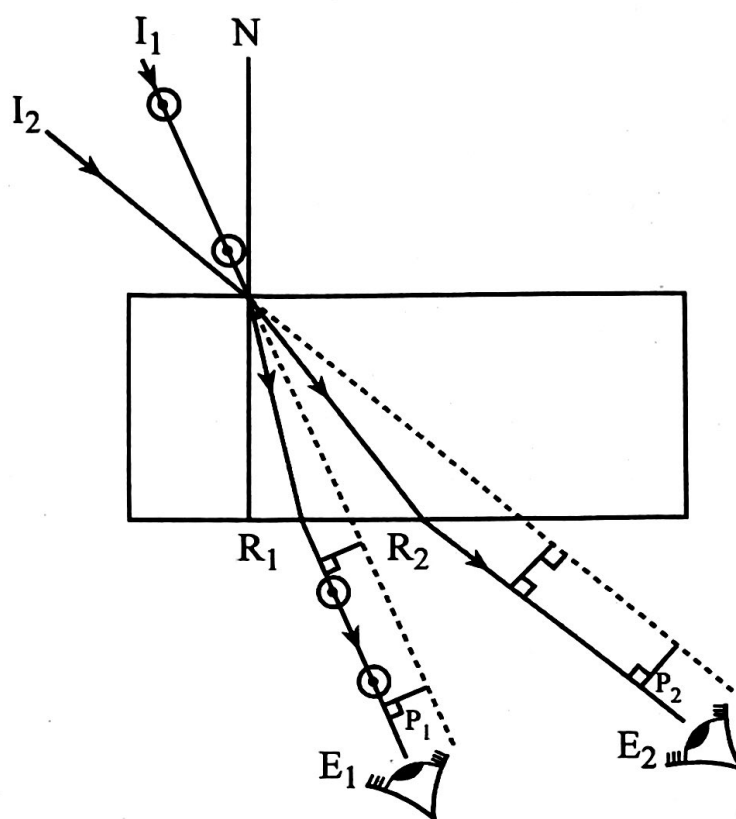
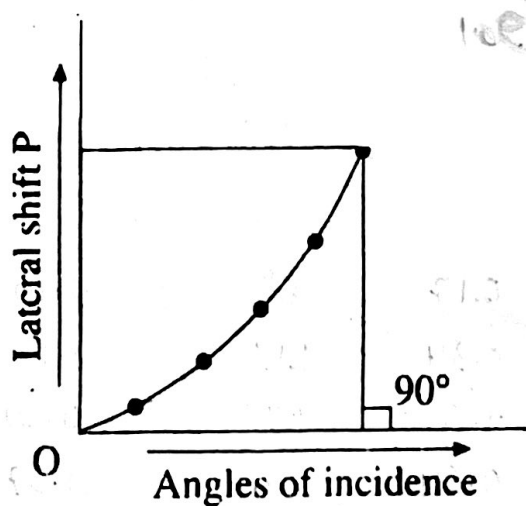
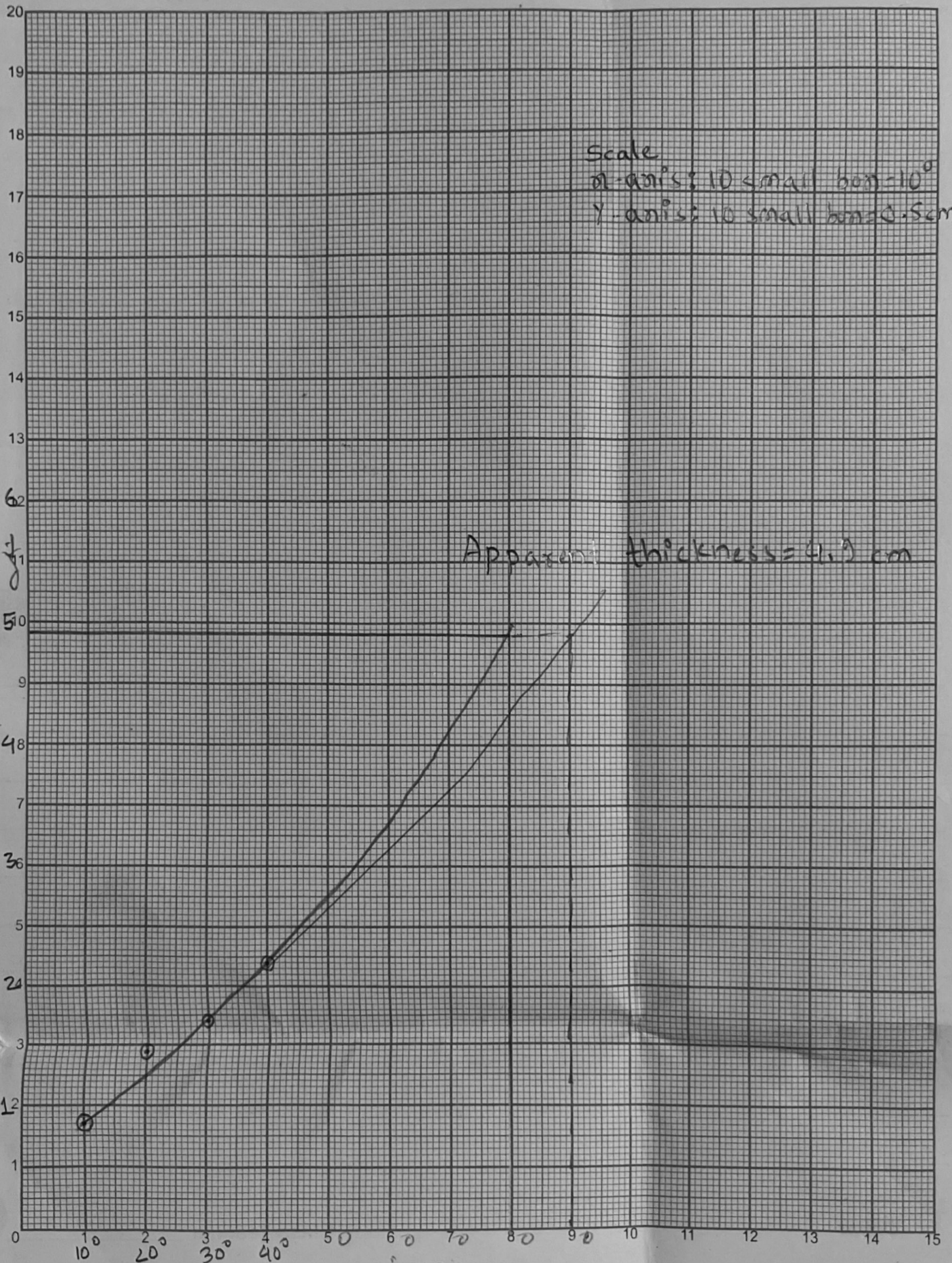


Fig: Refraction of light through glass slab at various angle of incidence





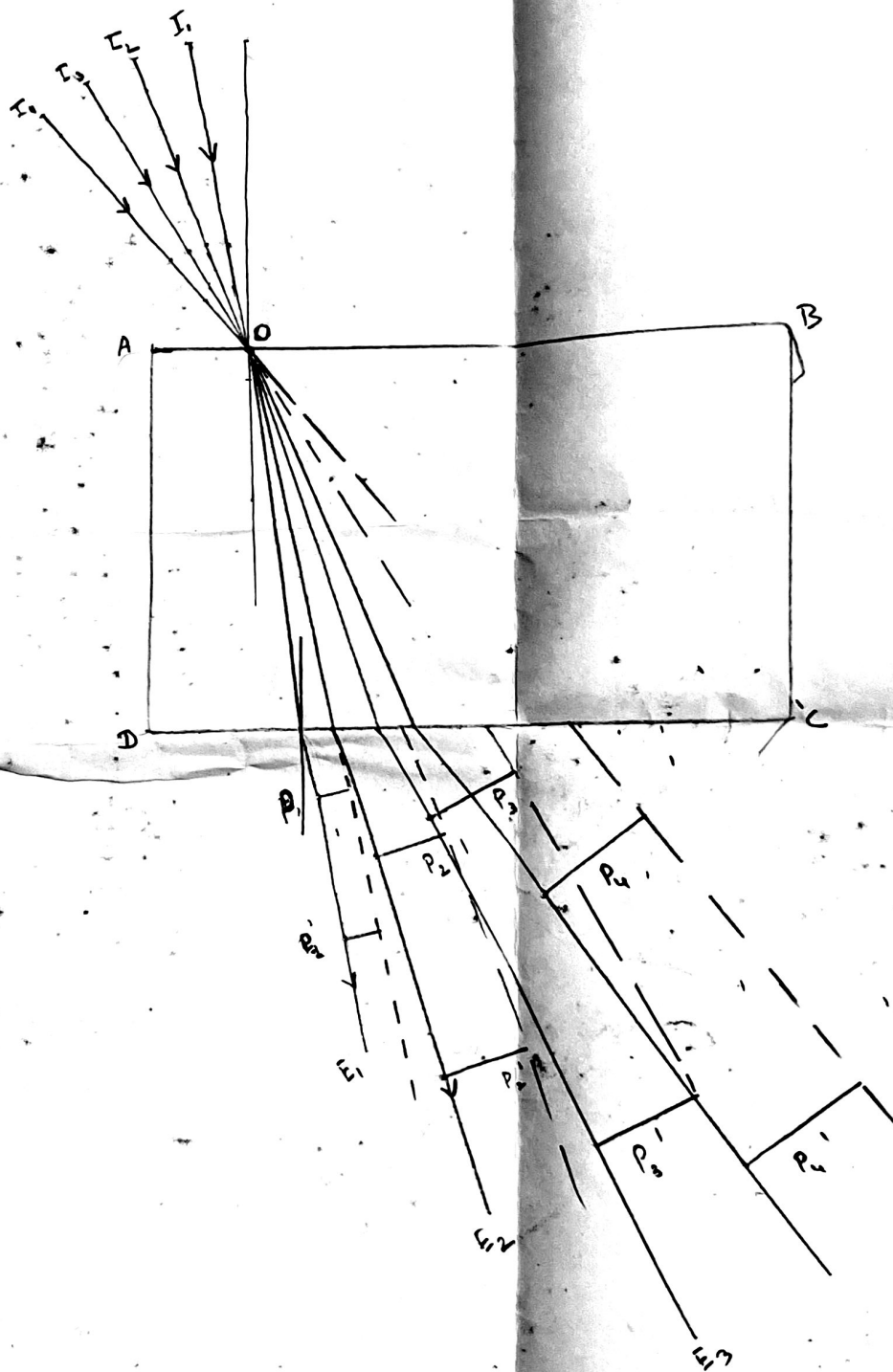


Table 2: For lateral shift

No. of Obs.	Angles of incidence (i)	Lateral Shift (P) cm
1	10°	0.85
2	20°	1.45
3	30°	1.7
4	40°	2.2

From graph, for $i = 90^\circ$, Lateral Shift (P) = thickness (t) = 2.2 cm

Actual thickness of the glass slab (measured by ruler) = 2.2 cm

RESULT:

From this experiment the variation of Lateral shift with angle of incidence is observed and from graph, Thickness of glass slab is found to be

CONCLUSION:

The laws of refraction were verified using plane mirror.

SOURCES OF ERROR:

The error may arise due to:-

- i. Shift of the glass slab from its original outline
- ii. Nonlinearity of object and image pins
- iii. Improper management of angle and length

PRECAUTIONS:

1. The glass slab should be its original outline
2. The separation between the pins should be about 10 cm.
3. Arrow head should be used to denote the direction of rays.

VIVA QUESTIONS:

1. Define lateral shift. Write its unit.
2. Is it possible to have zero lateral shift?
3. State Snell's law.
4. A ray of light passing from one medium to other medium bends. Why?
5. What is the condition of which lateral shift is equal to the thickness of the glass slab?
6. Define refractive index and write down its unit and dimension.
7. The length of a day increases because of refraction of light. Explain why?
8. What is total internal reflection?
9. Why does the partially immersed rod in water appear bent?

1. The perpendicular distance between the direction of incident ray and the emergent ray when a ray of light refracts from parallel edged glass slab, is called lateral shift.
Its unit is cm or m.

2. Yes, it is possible when the angle of incidence makes an angle of 0° with point of contact or is parallel with the normal.

3. Snell's law of refraction states that the ratio of sine angle of incidence to sine angle of refraction (at an interface between two optical media) is always constant.

$$\text{Mathematically, } \mu = \frac{\sin i}{\sin r}$$

4. Because the speed of light changes when it enters a different medium. This change in speed causes refraction (bending of light).

5. Lateral shift becomes equal to the thickness of the glass slab when the angle of incidence is 90° . (i.e. The ray grazes along the surface).

6. Refractive index (μ) is the ratio of the speed of light in vacuum to the speed of light in a medium.

$$\text{i.e. } \mu = \frac{\text{Speed of light in air (c)}}{\text{speed of light in medium (v)}}$$

It doesn't have unit and dimension.

7. Due to atmospheric refraction, we can see the sun before actual sunrise and after actual sunset. The atmosphere bends sunlight towards Earth, making the sun appear higher than its real position. Hence, the day appears about 4 minutes longer.

8. Total internal reflection is the complete reflection of light back into the denser medium when it travels from denser to rarer medium and the angle of incidence is greater than the critical angle.

9. Because of refraction of light at the water-air boundary. Light from the submerged part bends when it comes into air, making the lower part appear raised and the rod look bent.

